UNCOVERING NATION-STATE THREATS: INVESTIGATING CALLISTO THREAT ACTOR GROUP WITH MISP FOR ENHANCED CYBERSECURITY

AN INVESTIGATIVE ANALYSIS

CONDUCTED BY

STANLEY EKECHUKWU

FOR

SECUREOPS CYBERSECURITY INC., EUROPE

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EXECUTIVE SUMMARY

This report presents a structured threat intelligence assessment of an espionage campaign attributed to the Callisto threat actor group, based on analysis of MISP Event 1032. The investigation identifies a coordinated, phishing-driven operation targeting government, military, policy, and research entities across Europe and the South Caucasus.

The analysis confirms the presence of three integrated infrastructure components: large-scale phishing domains designed to impersonate trusted service providers; deployment artifacts linked to RCS Galileo spyware; and a dedicated command-and-control (C2) server used for remote access and data exfiltration. These elements collectively demonstrate a multi-stage intrusion lifecycle beginning with credential harvesting and, where necessary, escalating to endpoint compromise and persistent surveillance.

Behavioural observations were mapped to the MITRE ATT&CK framework to standardize analysis and align findings with recognized defensive models. The overall assessment indicates a strategically motivated espionage operation with moderate-to-high operational maturity.

INTRODUCTION

This project examines a documented Callisto espionage campaign using structured threat intelligence analysis derived from MISP Event 1032. The objective is to transform raw technical indicators into a coherent threat actor profile aligned with professional Security Operations Centre (SOC) and Cyber Threat Intelligence (CTI) methodologies.

The investigation addresses three core analytical tasks:

- Systematic extraction and organization of Indicators of Compromise (IOCs), including phishing domains, malware hashes, persistence artifacts, and command-and-control infrastructure.
- Categorization and analysis of infrastructure to determine operational roles within the attack lifecycle.
- Development of a comprehensive threat actor profile incorporating attribution, motivation, targeting patterns, and mapped Tactics, Techniques, and Procedures (TTPs).

The resulting assessment provides both technical clarity and strategic context to support defensive planning and organizational risk awareness.

PROJECT SCOPE

- Investigating the threat actor group Callisto and its infrastructure.
- Analysing threat intelligence data using MISP feeds.
- Reporting on the threat actor's TTPs and creating actionable intelligence for client protection.

TOOLS AND ENVIRONMENT

MISP Platform: Primary tool for sharing and analysing threat intelligence.

- Docker: Containerization for deploying MISP.
- Python: For automating IOC collection and analysis from MISP feeds.
- Elasticsearch: For indexing and searching large sets of threat data.
- Kali Linux: Operating system for conducting threat investigations and using security tools.

METHODOLOGY

The investigation followed a structured intelligence analysis workflow consistent with recognized CTI practices.

Data Collection

MISP Event 1032 was accessed and all associated attributes were exported in CSV format to ensure complete capture of technical artifacts, including domains, IP addresses, file hashes, filenames, and embedded analytical references.

Data Normalisation and Structuring

The exported dataset was cleaned and standardized to improve analytical clarity. This process included:

- Normalisation of indicator types
- Conversion of timestamps to readable formats
- Removal of non-essential metadata
- Validation of each IOC against event commentary

The refined dataset formed the technical foundation for subsequent analysis.

Infrastructure Categorisation

Indicators were grouped into functional infrastructure categories:

- Phishing infrastructure
- Malware delivery and persistence artifacts
- Command-and-control infrastructure

Each category was analysed to determine operational purpose within the intrusion chain.

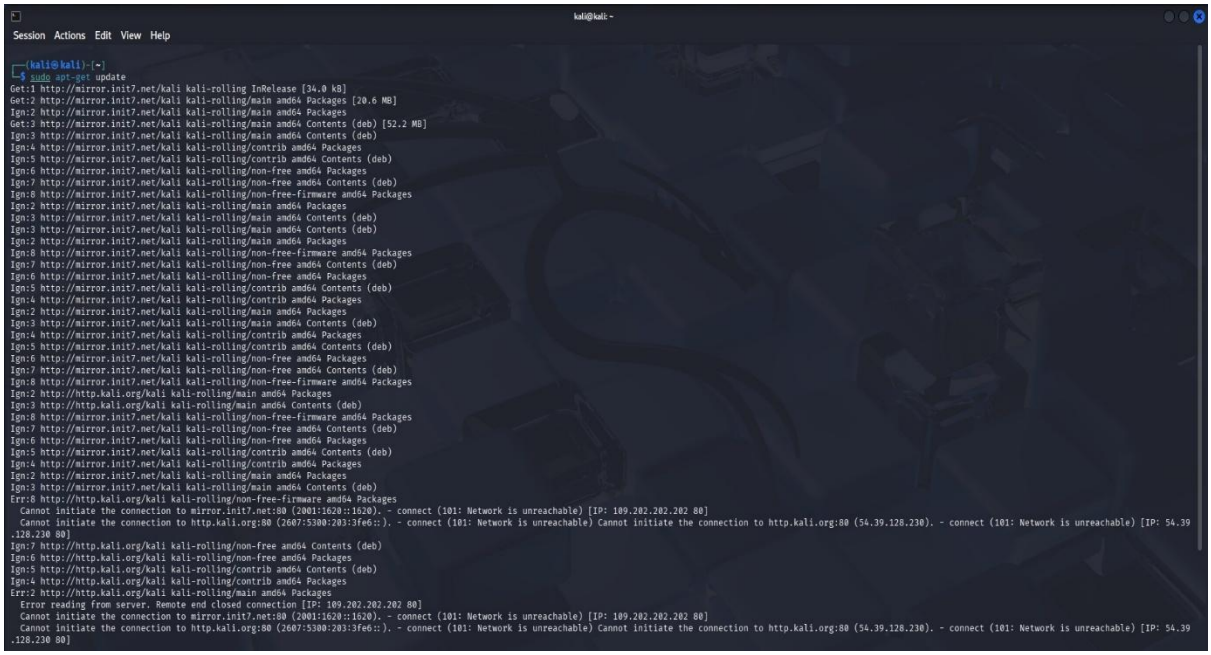
Behavioural Analysis and Framework Mapping

Observed behaviours were mapped to the MITRE ATT&CK framework to standardise TTP classification and align findings with defensive detection strategies. This ensured analytical consistency and industry alignment.

Validation and Quality Assurance

All conclusions were cross-checked against event comments and embedded references to ensure accuracy and evidence-based reporting. The assessment was conducted strictly within the scope of the provided dataset.

Kali Linux Configurations



Kali Linux Configurations Contd.

```
┌──(kali㉿kali)-[~]
└─$ sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
[sudo] password for kali:
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
docker-ce is already the newest version (5:29.2.1-1~debian.12~bookworm).
docker-ce-cli is already the newest version (5:29.2.1-1~debian.12~bookworm).
containerd.io is already the newest version (2.2.1-1~debian.12~bookworm).
docker-buildx-plugin is already the newest version (0.31.1-1~debian.12~bookworm).
docker-compose-plugin is already the newest version (5.0.2-1~debian.12~bookworm).
Solving dependencies ... Done
0 upgraded, 0 newly installed, 0 to remove and 1561 not upgraded.

┌──(kali㉿kali)-[~]
└─$ sudo apt install git
git is already the newest version (1:2.51.0-1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1561

┌──(kali㉿kali)-[~]
└─$ git clone https://github.com/MISP/misp-docker
Cloning into 'misp-docker'...
remote: Enumerating objects: 3002, done.
remote: Counting objects: 100% (1034/1034), done.
remote: Compressing objects: 100% (375/375), done.
remote: Total 3002 (delta 908), reused 659 (delta 659), pack-reused 1968 (from 2)
Receiving objects: 100% (3002/3002), 885.90 KiB | 13.00 KiB/s, done.
Resolving deltas: 100% (1616/1616), done.

┌──(kali㉿kali)-[~]
└─$ cd misp-docker/

┌──(kali㉿kali)-[~/misp-docker]
└─$ cp template.env .env

┌──(kali㉿kali)-[~/misp-docker]
└─$
```

```
WARN[0000] The "AAD_CLIENT_SECRET" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_REDIRECT_URI" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_PROVIDER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_PROVIDER_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_ORGADMIN" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_SITEADMIN" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_CHECK_GROUPS" variable is not set. Defaulting to a blank string.
WARN[0000] The "NGINX_X_FORWARDED_FOR" variable is not set. Defaulting to a blank string.
WARN[0000] The "NGINX_SET_REAL_IP_FROM" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_ENABLE" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_HOST" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_PORT" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_METHOD" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_BUCKET" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_ENDPOINT" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_ACCESS_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_SECRET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "STUNNEL" variable is not set. Defaulting to a blank string.
WARN[0000] The "STUNNEL_CONFIG" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_HOST" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_USERNAME" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_CA" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_CERT" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "DEBUG" variable is not set. Defaulting to a blank string.
WARN[0000] The "SMTP_FQDN" variable is not set. Defaulting to a blank string.
WARN[0000] The "FASTCGI_STATUS_LISTEN" variable is not set. Defaulting to a blank string.
WARN[0000] The "PHP_SESSION_COOKIE_DOMAIN" variable is not set. Defaulting to a blank string.
WARN[0000] The "HSTS_MAX_AGE" variable is not set. Defaulting to a blank string.
WARN[0000] The "X_FRAME_OPTIONS" variable is not set. Defaulting to a blank string.
WARN[0000] The "CONTENT_SECURITY_POLICY" variable is not set. Defaulting to a blank string.
WARN[0000] The "COMPOSE_PROFILES" variable is not set. Defaulting to a blank string.
WARN[0000] The "MODULES_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "GUARD_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "GUARD_ARGS" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_REGION" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PORT" variable is not set. Defaulting to a blank string.
[+] up 4/4
 ✔ Container misp-docker-db-1            Healthy                                                                    0.5s
 ✔ Container misp-docker-misp-modules-1  Healthy                                                                    0.5s
 ✔ Container misp-docker-mail-1          Running                                                                    0.0s
 ✔ Container misp-docker-redis-1         Healthy                                                                    0.5s

┌──(kali㉿kali)-[~/misp-docker]
└─$
```

```
WARN[0000] The "AAD_REDIRECT_URI" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_PROVIDER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_PROVIDER_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_ORGADMIN" variable is not set. Defaulting to a blank string.
WARN[0000] The "AAD_MISP_SITEADMIN" variable is not set. Defaulting to a blank string.
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WARN[0000] The "PROXY_PORT" variable is not set. Defaulting to a blank string.
WARN[0000] The "PROXY_METHOD" variable is not set. Defaulting to a blank string.
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WARN[0000] The "PROXY_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_BUCKET" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_ENDPOINT" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_ACCESS_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "S3_SECRET_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "STUNNEL" variable is not set. Defaulting to a blank string.
WARN[0000] The "STUNNEL_CONFIG" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_HOST" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_USERNAME" variable is not set. Defaulting to a blank string.
WARN[0000] The "SUPERVISOR_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_CA" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_CERT" variable is not set. Defaulting to a blank string.
WARN[0000] The "MYSQL_TLS_KEY" variable is not set. Defaulting to a blank string.
WARN[0000] The "DEBUG" variable is not set. Defaulting to a blank string.
WARN[0000] The "SMTP_FQDN" variable is not set. Defaulting to a blank string.
WARN[0000] The "FASTCGI_STATUS_LISTEN" variable is not set. Defaulting to a blank string.
WARN[0000] The "PHP_SESSION_COOKIE_DOMAIN" variable is not set. Defaulting to a blank string.
WARN[0000] The "HSTS_MAX_AGE" variable is not set. Defaulting to a blank string.
WARN[0000] The "X_FRAME_OPTIONS" variable is not set. Defaulting to a blank string.
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WARN[0000] The "COMPOSE_PROFILES" variable is not set. Defaulting to a blank string.
WARN[0000] The "MODULES_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "GUARD_COMMIT" variable is not set. Defaulting to a blank string.
WARN[0000] The "GUARD_ARGS" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_USER" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PASSWORD" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_REGION" variable is not set. Defaulting to a blank string.
WARN[0000] The "SES_PORT" variable is not set. Defaulting to a blank string.
[+] pull 5/5
 ✔ Image valkey/valkey:7.2                        Pulled                                                            1.9s
 ✔ Image mariadb:10.11                            Pulled                                                            2.5s
 ✔ Image ghcr.io/misp/misp-docker/misp-modules:latest Pulled                                                        1.2s
 ✔ Image ghcr.io/misp/misp-docker/misp-core:latest    Pulled                                                        1.3s
 ✔ Image ghcr.io/egos-tech/smtp:1.1.3             Pulled                                                            1.2s

┌──(kali㉿kali)-[~/misp-docker]
└─$
```
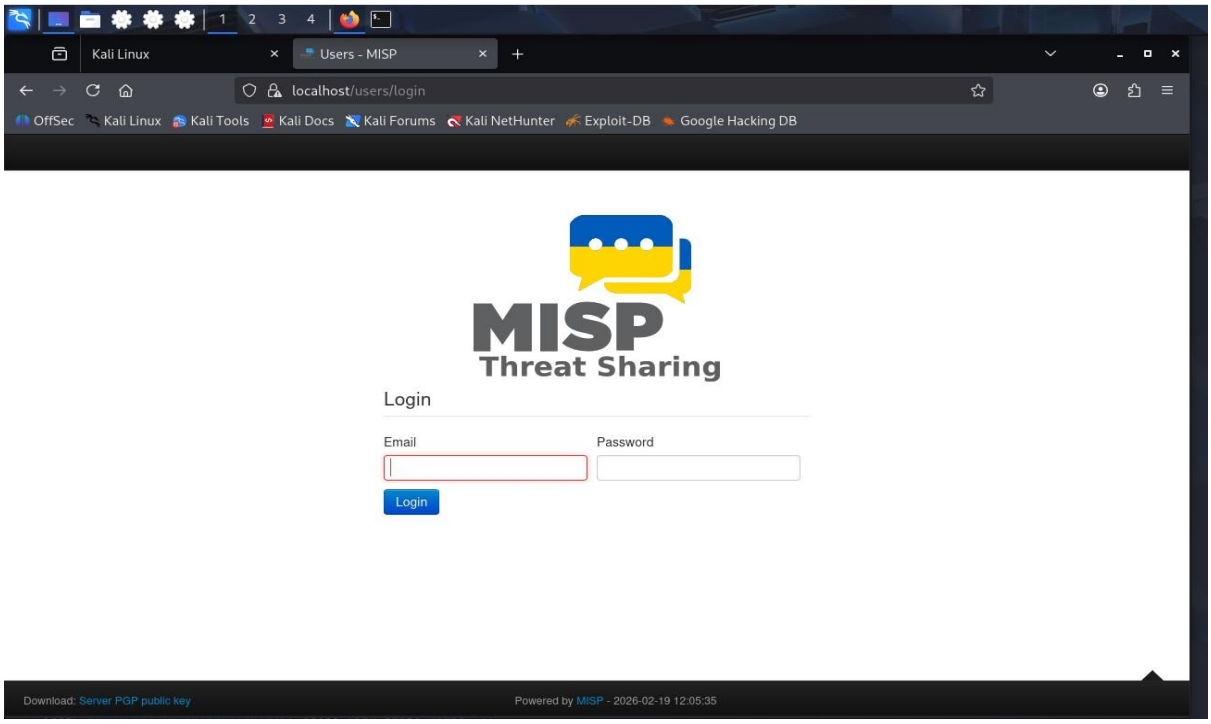
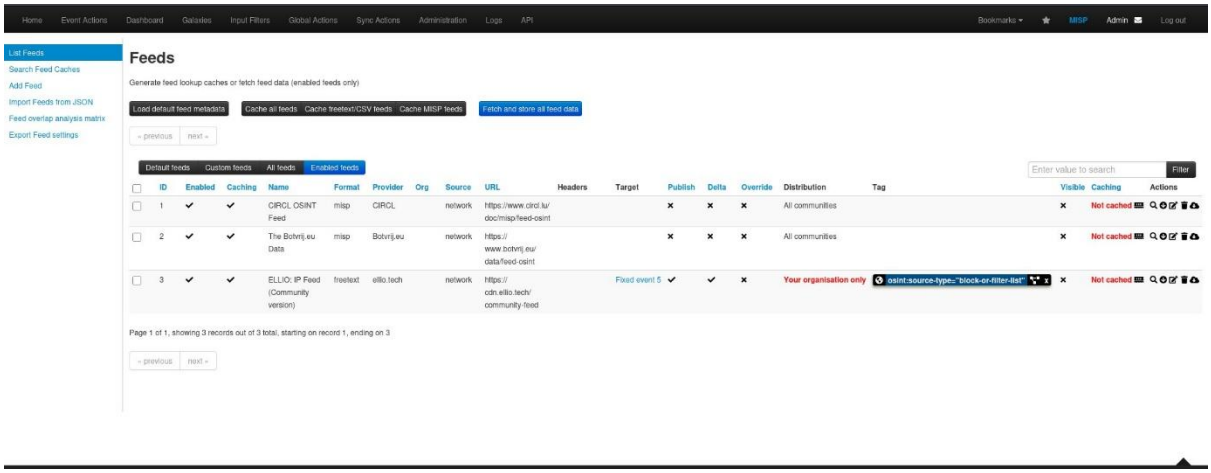
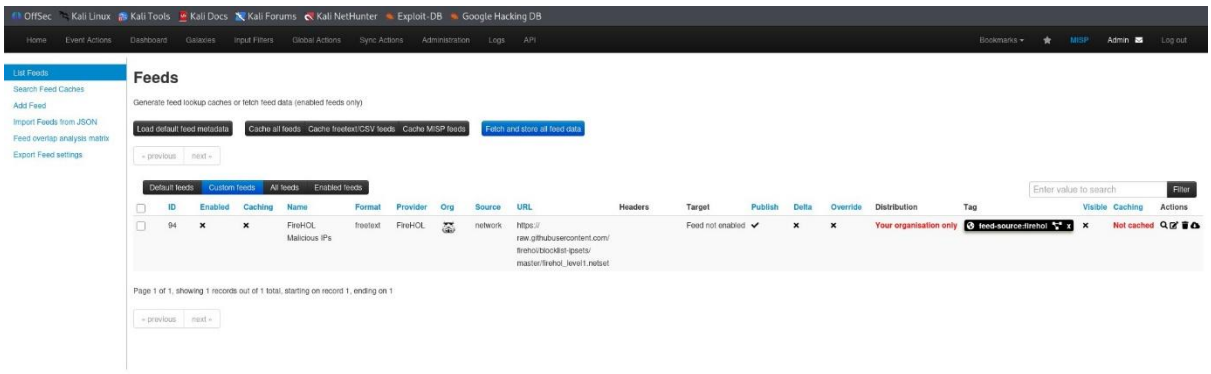

MISP login page



Enabled and Cached Feeds



Custom Feeds: Firehol

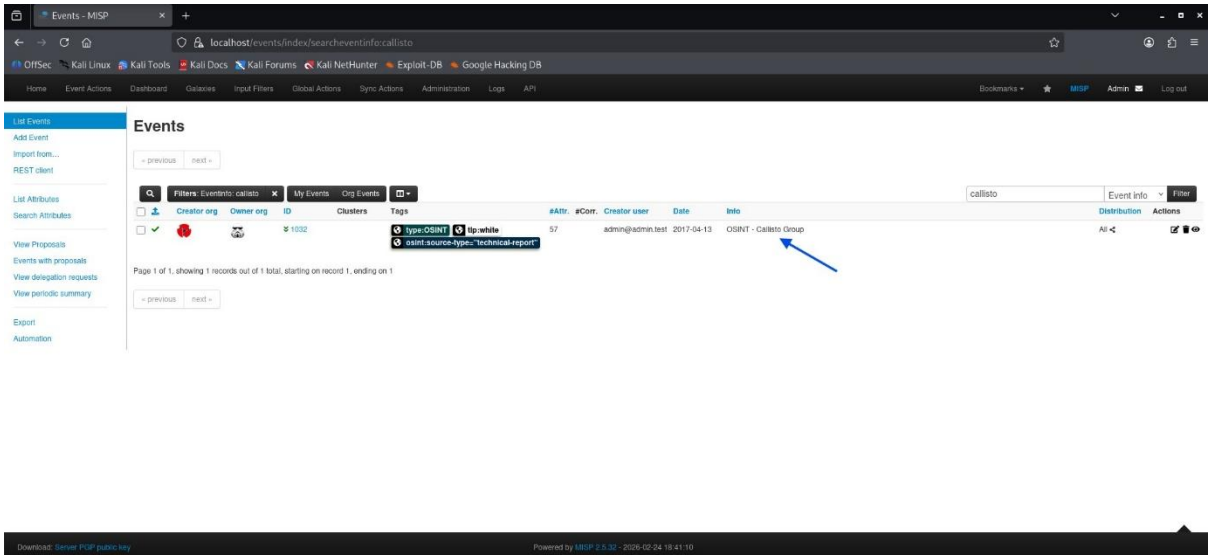


THREAT ACTOR GROUP (CALLISTO) INVESTIGATION

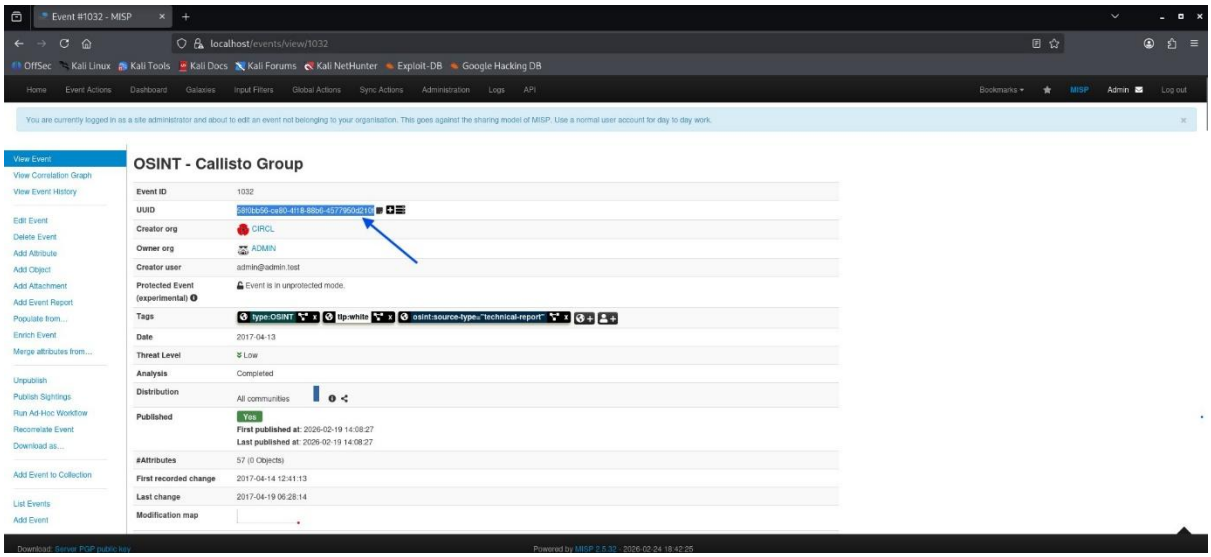
Custom Feed UUID - 5c415f76-1afc-45cc-96e3-01b2b735f4a6
Custom Feed Event ID - 1866
Callisto UUID - 58f0bb56-ce80-4f18-88b6-4577950d210f
Callisto Event ID - 1032

The Callisto group is an advanced threat actor whose targets include military personnel, government officials, think tanks, and journalists in Europe and the South Caucasus. Their primary interest appears to be gathering intelligence related to foreign and security policy in the Eastern Europe and South Caucasus regions.

Callisto Threat Group in MISP



Callisto UUID



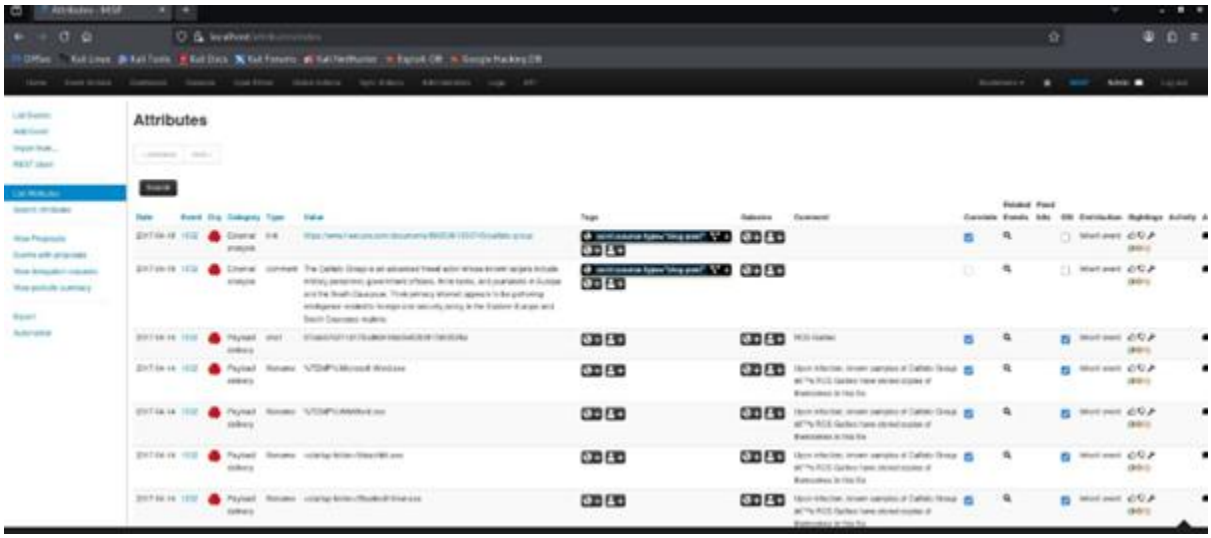
IOC Collection

Summary of Collected Indicators

From MISP Event 1032, the following IOCs were extracted:

- **Phishing domains:** 46
- **Malware hashes:** SHA-1, SHA-256, MD5 (all RCS Galileo)
- **Filenames:** 4 persistence artefacts
- **C2 IP address:** 1
- **External analysis links:** 2
- **Malware name:** RCS Galileo

Callisto IOCs as seen in MISP



All indicators were cleaned, normalised, and compiled into a structured IOC table.

		CALLISTO IOCs			
S/N	Indicator Type	IOC Value	Threat Actor	Severity	Timestamp
1	link	https://www.f-secure.com/documents/996508/1030745/callisto-group	Callisto	Low	19/04/2017
2	link	ttps://www.virustotal.com/file/974f6ceebeb889bd97e6641100dddf823376561ddde9e4749f3ea3d77f63a8f9/analysis/1492108895	Callisto	Low	14/04/2017
3	sha1	07cdc67d211d175cd9d418dc5482b3f17d93526a	Callisto	High	14/04/2017
4	sha256	974f6ceebeb889bd97e6641100dddf823376561ddde9e4749f3ea3d77f63a8f9	Callisto	High	14/04/2017
5	md5	99a18bf3c04a491b256f7d60eb6e0f26	Callisto	High	14/04/2017
6	ip-dst	89.46.102.43	Callisto	High	14/04/2017
7	malware-name	RCS Galileo	Callisto	High	14/04/2017
8	filename	%TEMP%\Microsoft Word.exe	Callisto	Medium	14/04/2017
9	filename	%TEMP%\WinWord.exe	Callisto	Medium	14/04/2017
10	filename	>startup folder<\bleachbit.exe	Callisto	Medium	14/04/2017
11	filename	>startup folder<\BluetoothView.exe	Callisto	Medium	14/04/2017
12	domain	accounts-google.eu	Callisto	High	14/04/2017
13	domain	accounts-mail.asia	Callisto	High	14/04/2017
14	domain	authentification-request.top	Callisto	High	14/04/2017
15	domain	auth-login.top	Callisto	High	14/04/2017
16	domain	drive-login.com	Callisto	High	14/04/2017
17	domain	drive-meet-goodle.ru	Callisto	High	14/04/2017
18	domain	emailapp.pw	Callisto	High	14/04/2017
19	domain	fco-gov.pw	Callisto	High	14/04/2017
20	domain	fco-net.pw	Callisto	High	14/04/2017
21	domain	google-plus.top	Callisto	High	14/04/2017
22	domain	google-accounts.eu	Callisto	High	14/04/2017
23	domain	google-service.eu	Callisto	High	14/04/2017
24	domain	hotmail-online.eu	Callisto	High	14/04/2017
25	domain	icloud-service.pw	Callisto	High	14/04/2017
26	domain	live-com.pw	Callisto	High	14/04/2017
27	domain	live-login.info	Callisto	High	14/04/2017
28	domain	login-access.top	Callisto	High	14/04/2017
29	domain	login-live.review	Callisto	High	14/04/2017
30	domain	login-livecom.in	Callisto	High	14/04/2017
31	domain	login-livecom.info	Callisto	High	14/04/2017

IOC Table contd.

31	domain	login-livecom.info	Callisto	High	14/04/2017
32	domain	login-live-com.pw	Callisto	High	14/04/2017
33	domain	misrcosofts.com	Callisto	High	14/04/2017
34	domain	node005-prevention-aol.link	Callisto	High	14/04/2017
35	domain	node03-prevention-icloud.link	Callisto	High	14/04/2017
36	domain	platforma.link	Callisto	High	14/04/2017
37	domain	prevention-aol.link	Callisto	High	14/04/2017
38	domain	prevention-aol.top	Callisto	High	14/04/2017
39	domain	prevention-icloud.link	Callisto	High	14/04/2017
40	domain	qooqle-support-mail.pw	Callisto	High	14/04/2017
41	domain	screenname.click	Callisto	High	14/04/2017
42	domain	screenname-aol.pw	Callisto	High	14/04/2017
43	domain	secure-lcloud.accountant	Callisto	High	14/04/2017
44	domain	secure-store-lcloud.top	Callisto	High	14/04/2017
45	domain	service-mail.asia	Callisto	High	14/04/2017
46	domain	service-mail.in	Callisto	High	14/04/2017
47	domain	serv-login-com.pw	Callisto	High	14/04/2017
48	domain	shared-docs.pw	Callisto	High	14/04/2017
49	domain	store-icloud.link	Callisto	High	14/04/2017
50	domain	support-gmail.pw	Callisto	High	14/04/2017
51	domain	support-mail.pw	Callisto	High	14/04/2017
52	domain	support-mail.top	Callisto	High	14/04/2017
53	domain	updatemail.in	Callisto	High	14/04/2017
54	domain	yahoocentermail.info	Callisto	High	14/04/2017
55	domain	yahoocentermail.pw	Callisto	High	14/04/2017
56	domain	yahoomailfree.pw	Callisto	High	14/04/2017
57	domain	go-veryfication.link	Callisto	High	14/04/2017

ANALYSIS AND REPORT ON IOCs

The indicators extracted from **Event 1032** represent a complete set of technical artefacts associated with a Callisto Group espionage campaign. The dataset includes phishing domains, a command-and-control (C2) IP address, malware hashes, persistence filenames, and external analysis references. Together, these indicators provide a clear view of the infrastructure and tooling used by Callisto to conduct credential theft and deploy spyware against targeted individuals in Europe and the South Caucasus.

Type of infrastructure in this context simply indicates the kind of technical resources the threat actor is using to run the campaign. The IOCs reveal **three (3)** main types of infrastructure used by Callisto:

1. Phishing Infrastructure (fake login domains): The domains are centred around impersonation of major service providers such as Google/Gmail, Microsoft/Hotmail, Apple/iCloud, Yahoo, AOL, and generic authentication/login lures. Some examples from the dataset include:

- accounts-google.eu
- auth-login.top
- drive-login.com
- fco-gov.pw
- yahoocentermail.pw
- icloud-service.pw
- login-live.review
- go-veryfication.link

This simply indicates that Callisto uses large-scale phishing infrastructure and the domains are crafted to mimic and impersonate major email/cloud providers and government-related entities. These domains are used in relation to phishing either as link targets or sender domains. This infrastructure hosts fake login pages, sends phishing emails, steals credentials, and impersonates legitimate organisations. This simply shows that Callisto's initial access vector is **credential harvesting**.

2. Malware Delivery Infrastructure (RCS Galileo hashes and persistence filenames): The IOC contains SHA-1, SHA-256, and MD5 hashes and refer to the same spyware – **RCS Galileo.**
Filenames used by the malware are:

- %TEMP%\Microsoft Word.exe
- %TEMP%\WinWord.exe
- >startup folder<\bleachbit.exe
- >startup folder<\BluetoothView.exe

This infrastructure delivers RCS Galileo spyware, stores copies of the spyware, masquerades as legitimate apps, and establishes persistence. This further confirms Callisto uses **stealthy spyware** to maintain long-term access.

3. Command-and-Control Infrastructure (C2 IP 89.46.102.43): Callisto uses this IP as a command and control (C2) server for this campaign. This is a high value IOC for detection and blocking as it represents remote endpoint used to control infected hosts or receive data. This IP is the only C2 node available in this dataset. These components collectively support a credential-harvesting and espionage campaign targeting European diplomatic and government entities. This infrastructure receives stolen data, sends commands to infected hosts, and maintains remote access.

Callisto is running a **phishing-driven espionage campaign** targeting military, government, think tanks, and journalists in Europe and the South Caucasus.

- The group uses a **large, themed phishing domain infrastructure** impersonating major providers (Google, Microsoft, Yahoo, Apple, AOL) and government-related entities.
- The campaign deploys **RCS Galileo** spyware, confirmed by SHA-1, SHA-256, and MD5 hashes and cross-checked via VirusTotal.
- RCS Galileo stores itself under **masquerading filenames** in %TEMP% and the **startup folder**, supporting stealth and persistence.
- A single **C2 IP (89.46.102.43)** is identified as a **"Known command & control server"**, providing a clear network IOC for detection and blocking.
- External links to F-Secure and VirusTotal validate that this is a **documented Callisto operation** focused on **intelligence gathering** related to foreign and security policy in Eastern Europe and the South Caucasus.

Correlation – No correlation based on MISP data

Within the scope of Event 1032, the provided IOCs form a self-contained phishing and spyware campaign attributed to the Callisto Group, no additional correlation data is present in this dataset. So, cross-event reuse of infrastructure cannot be assessed from this dataset alone.

THREAT ACTOR PROFILE: CALLISTO

Overview

Callisto is a well-documented espionage-focused threat actor targeting military personnel, government officials, think tanks, journalists, and policy-focused institutions across Europe and the South Caucasus. The group's operations are centred on intelligence collection related to foreign and security policy matters.

Analysis of MISP Event 1032 confirms a structured phishing-driven campaign supported by credential harvesting infrastructure, selective deployment of spyware (RCS Galileo), and a dedicated command-and-control (C2) server.

Attribution & Identity

Known Aliases:
BlueCharlie, COLDRIVER, SEABORGIUM, Star Blizzard, TA446

Suspected Origin:
Public intelligence reporting attributes Callisto to a Russia-aligned threat cluster (medium-to-high confidence).

Motivation:
Strategic espionage and long-term intelligence collection.

Primary Targets:

- Government and diplomatic entities
- Military personnel
- Think tanks (research organisations) and academia
- Journalists and policy researchers

Tools/Malware used: Infrastructure Model

The campaign observed in Event 1032 consists of **three** coordinated infrastructure components:

I. **Phishing Infrastructure**
Over 40 domains impersonating major service providers (Google, Microsoft, Apple, Yahoo) and government-related entities to harvest credentials.

II. **Malware Delivery & Persistence**
Cryptographic hashes confirm deployment of **RCS Galileo spyware**, stored using masquerading filenames:

- %TEMP%\Microsoft Word.exe
- %TEMP%\WinWord.exe
- >startup folder<\bleachbit.exe
- >startup folder<\BluetoothView.exe

These artifacts indicate stealth execution and startup persistence.

III. **Command-and-Control Infrastructure**
IP address **89.46.102.43** identified as a **dedicated C2 server** used for remote tasking and data exfiltration.

This structure reflects a multi-stage intrusion lifecycle beginning with credential harvesting and escalating to device-level compromise where necessary.

Observed Tactics, Techniques, and Procedures (TTPs)

Based on Event 1032, Callisto demonstrates the following behavioural patterns:

- Credential harvesting through impersonation domains
- User-driven execution via phishing links
- Deployment of spyware for sustained access
- Persistence through startup folder execution
- Masquerading of malicious executables as legitimate software
- Use of dedicated C2 infrastructure for remote control
- Email and document intelligence collection

MITRE ATT&CK Mapping

The observed behaviours were mapped to the MITRE ATT&CK framework to standardise analysis and align findings with defensive detection models.

ATT&CK Tactic	Technique	Technique ID	Evidence from Event 1032
Initial Access	Spearphishing Link	T1566.002	46 impersonation domains
Execution	User Execution	T1204	Victim interaction required
Persistence	Boot or Logon Autostart – Startup Folder	T1547.001	Startup folder executables
Defence Evasion	Masquerading	T1036	Fake Microsoft Word filenames
Command & Control	Application Layer Protocol	T1071	Dedicated C2 IP (89.46.102.43)
Collection	Email Collection	T1114	Targeting of email accounts for intelligence

This mapping confirms a phishing-driven intrusion model with selective endpoint compromise capabilities, consistent with an espionage-focused threat actor.

MITIGATION AND RECOMMENDATIONS

The Callisto campaign demonstrates a phishing-led intrusion model supported by credential harvesting, selective spyware deployment, and structured command-and-control communication. Defensive measures should therefore prioritise identity protection, phishing disruption, endpoint monitoring, and outbound traffic visibility.

1. Enforce Strong Identity Controls

- Implement mandatory Multi-Factor Authentication (MFA) across email, cloud, VPN, and privileged accounts.
- Enable conditional access policies and anomalous login detection (e.g., impossible travel, high-risk sign-ins).

2. Strengthen Email and Domain Security

- Deploy advanced anti-phishing and URL inspection controls.
- Enforce SPF, DKIM, and DMARC policies.
- Implement DNS filtering to block impersonation and malicious domains.

3. Enhance Endpoint Monitoring and Persistence Detection

- Deploy Endpoint Detection and Response (EDR) solutions.
- Monitor for executables running from temporary directories.
- Alert on unauthorised modifications to startup folders.
- Detect masquerading filenames mimicking legitimate applications.

4. Monitor and Restrict Command-and-Control Communication

- Establish baseline outbound network traffic.
- Alert on connections to unknown or suspicious external IP addresses.
- Integrate threat intelligence feeds into firewall and IDS/IPS controls.

5. Implement Targeted Security Awareness

Provide focused training for high-risk personnel on:

- Identifying phishing emails.
- Recognising impersonation domains.
- Reporting suspicious login prompts promptly.

CONCLUSION

This investigation provides a structured threat intelligence assessment of the Callisto threat actor based on analysis of MISP Event 1032. Through systematic extraction and categorisation of indicators, the campaign was deconstructed into its core operational components: phishing-based credential harvesting infrastructure, selective spyware deployment with persistence mechanisms, and dedicated command-and-control capability.

The findings confirm that Callisto operates primarily as an espionage-focused actor leveraging socially engineered initial access rather than direct exploitation. The large volume of impersonation domains demonstrates deliberate infrastructure staging designed to target government, military, and policy-focused individuals. The presence of spyware artifacts and startup persistence mechanisms indicates the capability to escalate from credential harvesting to sustained endpoint surveillance when required.

Mapping observed behaviours to the MITRE ATT&CK framework further validates the structured nature of the campaign and aligns the analysis with industry-recognised threat modelling standards. This approach enables defensive teams to translate technical findings into actionable detection and mitigation strategies.

Overall, the assessed campaign reflects moderate-to-high operational maturity characterised by targeted victim selection, layered infrastructure, and strategic intelligence collection objectives. Organisations operating within government, defence, and policy sectors remain at elevated risk and must prioritise identity security, phishing detection, endpoint monitoring, and outbound traffic visibility to reduce exposure.

This investigation demonstrates the value of structured threat intelligence analysis in transforming raw indicators into operational insight capable of informing both technical defences and strategic risk management decisions.